CHEMISTRY ONLY

9 METALLIC BONDING

In this chapter we will discuss the bonding in metals and explain some of the physical properties of metals in terms of structure and bonding.



▲ Figure 9.1 Metals have properties that make them useful as construction materials.



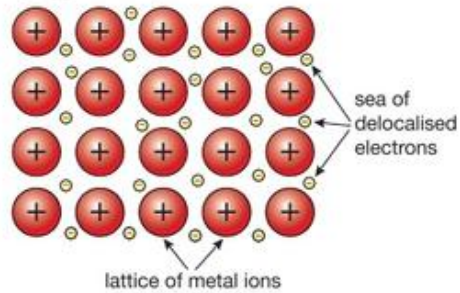
▲ Figure 9.2 Most metals are strong and all metals conduct electricity.

LEARNING OBJECTIVES

- Know how to represent a metallic lattice by a 2D diagram
- Understand metallic bonding in terms of electrostatic attractions
- Explain typical physical properties of metals, including electrical conductivity and malleability

METALLIC BONDING

Sodium is a metal. When sodium atoms bond together to form the solid metal, the outer electron on each sodium atom becomes free to move throughout the whole structure. The electrons are said to be *delocalised*. These electrons are no longer attached to particular atoms or pairs of atoms, instead you can think of them as flowing around through the whole metal. When a sodium atom loses its outer electron a sodium ion (Na^+) is left behind. A metallic structure consists of *a lattice (regular arrangement) of positive ions in a sea of delocalised electrons* (Figure 9.3).



▲ Figure 9.3 The metallic structure of a metal such as sodium

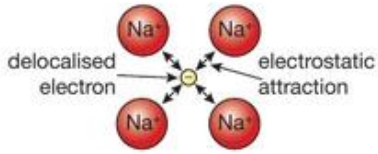
HINT

Different metals have different arrangements of ions in the lattice. Do not worry too much about this when drawing a diagram of a metal. As long as you draw the ions in a regular arrangement that will be fine.

HINT

When they come to write the symbol for a metal such as sodium in equations, students who know about metallic bonding sometimes worry whether they should write it as Na or Na^+. You write it as atoms, as Na. Thinking about the structure as a whole, the number of electrons exactly balances the number of positive charges. The metal as a whole carries no charge.

Metallic bonding is the electrostatic forces of attraction between each positive ion and the delocalised electrons. This holds the structure together.



▲ Figure 9.4 The electrostatic attraction between positively charged ions and negatively charged electrons holds the metallic structure together.

Metals have giant structures. There are no individual molecules and all the positive ions in the lattice attract all the delocalised electrons.

The ion formed by the metal depends on the number of electrons the original atom has in its outer shell. Thus all the elements in Group 1 form 1+ ions and all the elements in Group 2 form 2+ ions.

PHYSICAL PROPERTIES OF METALS

Most metals are hard and have high melting points. This suggests that the electrostatic forces of attraction between the positive ions and the delocalised electrons are strong.

In the case of sodium, only one electron per atom is delocalised, leaving ions with only one positive charge on them. The bonding in sodium is quite weak, as metals go, which is why sodium is fairly soft, with a low melting point for a metal. Magnesium has two outer electrons, both of which are delocalised into the 'sea', leaving behind ions that carry a charge of 2+. There is a much stronger electrostatic attraction between the 2+ ions and the delocalised electrons. This means that the bonding is stronger in magnesium and the melting point is higher.

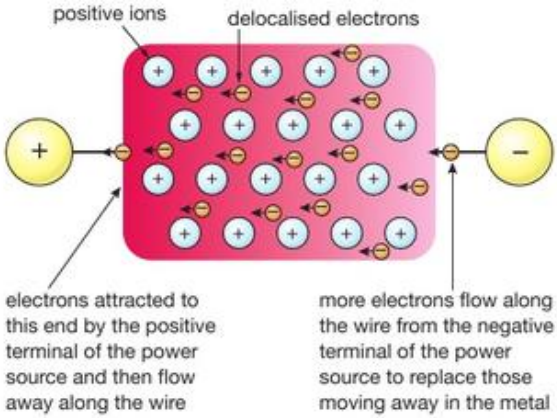
METALS CONDUCT ELECTRICITY

KEY POINT

In Chapter 7 we discussed ionic compounds and how they conduct electricity when molten or in aqueous solution because the **ions** are free to move. Ionic substances do not conduct electricity when solid because the ions are not free to move.
Metals conduct electricity when solid or molten because the delocalised **electrons** are free to move. It is important not to confuse this:

- ionic substances: ions move
- metals: electrons move

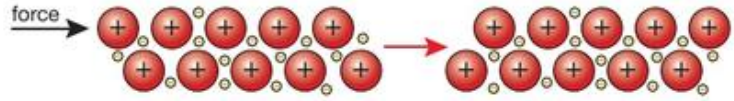
Metals conduct electricity because the *delocalised electrons are free to move throughout the structure*. Imagine what happens if a piece of metal is attached to an electrical power source.



▲ Figure 9.5 How metals conduct electricity.

METALS ARE MALLEABLE

Metals can be hammered into different shapes; this is what the word **malleable** means. When we apply a force to a piece of metal *the layers of positive ions slide over each other*. This does not affect the bonding in the structure; the positive ions are still attracted to the delocalised electrons.



▲ Figure 9.6 Metals are malleable because the layers of positive ions can slide over each other without changing the bonding.

Metals can also be described as **ductile**. This means that they can be drawn out into wires. The explanation is the same as why they are malleable.

CHAPTER QUESTIONS

SKILLS INTERPRETATION 7

1 Draw simple 2D diagrams to show the structure of:
- a potassium metal
- b calcium metal

SKILLS CRITICAL THINKING 8

SKILLS REASONING 10

2 a A solid metal is often described as having 'a lattice of positive ions in a sea of delocalised electrons'. State the electronic configuration of a magnesium atom and use it to explain what this phrase means.

b Metallic bonds are not fully broken until the metal has first melted and then boiled. Sodium, magnesium and aluminium are three consecutive elements in the Periodic Table. The boiling points of sodium, magnesium and aluminium are 890 °C, 1110 °C and 2470 °C, respectively. Explain these values in terms of the electronic configurations of the elements and metallic bonding.

7 c Explain why all these elements are good conductors of electricity.

d Explain why all these metals are malleable.

9 3 This question uses ideas from Chapters 7, 8 and 9. In these chapters you have met the following types of structure and bonding:

giant metallic structure
giant covalent structure
giant ionic structure
covalent molecular structure

Some information about some substances is given below. In each case state what type of structure and bonding it has.

a Substance A melts at 2300 °C. It doesn't conduct electricity even when it is molten. It is insoluble in water.

8 b Substance B is a colourless gas.

9 c Substance C is a yellow solid with a low melting point of 113 °C. It doesn't conduct electricity and it is insoluble in water.

d Substance D forms brittle orange crystals which melt at 398 °C. It dissolves freely in water to give an orange solution.

e Substance E is a pink-brown flexible solid. It conducts electricity.

f Substance F is a liquid with a boiling point of 80 °C. It is insoluble in water.

g Substance G is a silvery solid which melts at 660 °C. It is used in overhead power cables.

END OF CHEMISTRY ONLY